WAIMLAp 2026

Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

Masked Linear Cross-Attention for Robust RGB-D Semantic Segmentation under Incomplete Depth

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WORK IN PROGRESS · Method and evaluation protocol defined; experiments under way. Results will be added to the final poster.

MOTIVATION & RESEARCH PROBLEM

- RGB-D sensors on robots, AR devices and mapping systems fail systematically: reflective, transparent, dark or distant surfaces leave **holes**, and edges produce **noise**.
- Most RGB-D segmentation methods fuse every depth position as if it were valid; missing values (zeros) leak into the fused features.
- Cross-modal attention (e.g. CMX [1]) improves fusion but is quadratic and has no notion of measurement validity.

Question: *Does excluding invalid depth inside a linear cross-attention reduce the accuracy loss when depth is noisy, missing or absent, at a small computational cost?*

OBJECTIVES / CONTRIBUTION

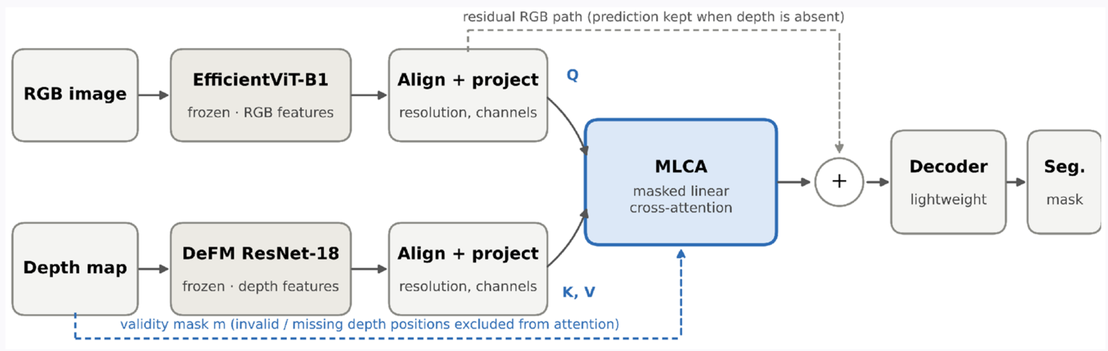
Main objective: measure how much validity masking inside linear cross-attention protects RGB-D segmentation from degraded depth.

- **C1 Method:** MLCA, a lightweight masked linear cross-attention for RGB + depth fusion.
- **C2 Experimental:** controlled ablations under partial, contiguous, noisy and total depth loss.
- **C3 Computational:** accuracy–robustness–cost trade-off (mIoU, params, FLOPs, latency, VRAM).

Novelty is combinatorial: the components exist; we isolate the effect of masking and will report it even if it does not help.

METHODOLOGY

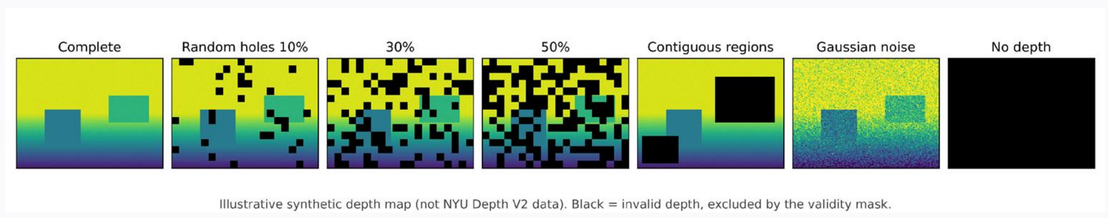
Frozen EfficientViT-B1 [2] extracts RGB features and frozen DeFM ResNet-18 [3], distilled from a depth-only foundation model, extracts geometric features. Only projections, MLCA and the decoder are trained.



$$MLCA(Q,K,V)_i = \varphi(Q_i)^T \Sigma_j m_j \varphi(K_j) V_j \;/\; (\varphi(Q_i)^T \Sigma_j m_j \varphi(K_j) + \varepsilon)$$

Kernelized linear attention [4] with RGB queries and depth keys/values. $m_j = 1$ if depth token j is valid; masked tokens leave both sums, so cost stays **O(N)** and the residual RGB path takes over when no depth is valid.

Degradation protocol



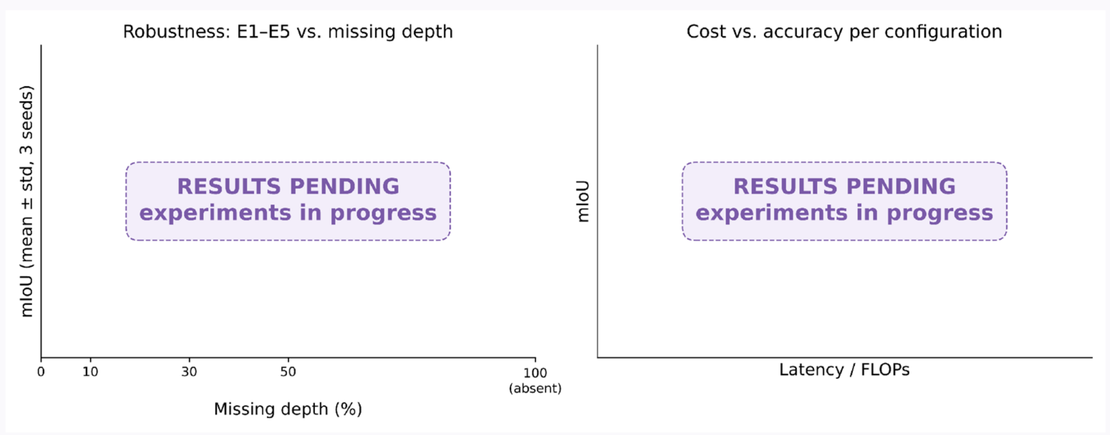
Data: NYU Depth V2 [5], 1,449 labeled indoor RGB-D images, 795 train / 654 test.
Metrics: mIoU, per-class IoU, mIoU drop vs. complete depth, params, FLOPs, latency, VRAM; mean ± std over 3 seeds.

RESULTS & DISCUSSION (IN PROGRESS)

Ablation design. Each step adds one component; the expected outcome is written before running it.

ID	Configuration	Isolates	Hypothesis (before running)
E1	EfficientViT-B1, RGB only	Baseline	Reference; unaffected by depth loss
E2	+ DeFM, concatenation	Adding depth	Gain with complete depth, drops as holes grow
E3	+ linear cross-attn, no mask	Attention fusion	Better fusion, similar sensitivity to holes
E4	+ validity mask (MLCA)	Masking	Smaller drop under holes; ≈ E1 with no depth; no gain under noise
E5	E4 ± modality dropout	Arch. vs. training	Dropout alone may match E4 → negative result

Planned evidence. The two plots below will hold the main results; no numbers are reported yet.



What would change our mind: if E4 ≈ E3 under holes, masking adds nothing; if E5 without the mask ≈ E4, robustness comes from training, not from the operator.

CONCLUSIONS

Main finding(s): MLCA is expected to reduce the mIoU drop relative to unmasked fusion when depth is incomplete or missing, and to stay close to RGB-only performance when depth is absent, with a bounded increase in parameters and latency.
Scientific / technological relevance: It would quantify the specific contribution of validity masking in linear attention and enable efficient RGB-D segmentation that is robust to sensor failures, useful for robotics and edge devices.
Current limitations: Incremental novelty (the components already exist), a binary mask that does not detect noisy depth, a small NYUv2 dataset, frozen extractors, and simulated failures
Next steps: Implement and evaluate E1 to E5 on NYUv2 under all degradation conditions (three seeds), confirm novelty in the literature, add a reliability estimate for noisy depth.

KEY REFERENCES

[1] J. Zhang et al., “CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers,” IEEE T-ITS, 2023.
[2] H. Cai et al., “EfficientViT: Lightweight multi-scale attention for high-resolution dense prediction,” ICCV, 2023.
[3] M. Patel et al., “DeFM: Learning foundation representations from depth for robotics,” arXiv:2601.18923, 2026.
[4] A. Katharopoulos et al., “Transformers are RNNs: Fast autoregressive transformers with linear attention,” ICML, 2020.
[5] N. Silberman et al., “Indoor segmentation and support inference from RGBD images,” ECCV, 2012.

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Github: https://github.com/joagovi/inf659-rgbd-mlca
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